Ps 03

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1 Problem 7

1.1 Task

(33) Discrete cosine transforms: multilevel structure

We continue the treatment of discrete cosine transforms. In this problem, you will practice: matrix notations (§ II.3.i) and some patterned matrices (Definition II.3.8) in parts (b), (e); permutation matrices in parts (b) and (f); linear operations on matrices and their properties (Definition II.3.4 and II.3.5, Proposition II.3.7) in part (j); matrix multiplication (Definition II.4.3 and Remark II.4.5) in parts (b), (d), (e), (h), (j); matrix transposition (Definition II.3.12) in parts (h) and (j); matrix inversion (Definition II.4.14) in parts (c) and (g); implementation in Julia in parts (k) and (l); trigonometry!

Pay attention to notational differences from Problem 3. Some DCTs of order $n \in \mathbb{N}$ involve the matrices $B, C \in \mathbb{R}^{n\times n}$ given by $B_{ij} = \cos[\frac{\pi}{n}(i-\frac{1}{2})(j-\frac{1}{2})]$ and $C_{ij} = \cos[\frac{\pi}{n}(i-\frac{1}{2})(j-1)]$ (1) for all $i, j \in \{1, ..., n\}$.

For $n \in \mathbb{N}$ we consider the matrices $B, C \in \mathbb{R}^{n\times n}$ given by (1) and the matrices $\hat{B}$, $\hat{C} \in \mathbb{R}^{2n\times 2n}$ defined analogously, with $2n$ in place of n in (1). Our goal is to relate the larger matrices $\hat{B}$ and $\hat{C}$ to the smaller matrices B and C. To this end, we select four submatrices of size $n \times n$ in each of $\hat{B}$ and $\hat{C}$ as follows. For all $\alpha, \beta \in \{1, 2\}$, the submatrices $\tilde{B}_{\alpha\beta}$ and $\tilde{C}_{\alpha\beta}$ (note Remark II.3.3 in the lecture notes) are formed by the first n (for $\alpha = 1$) or by the last n ($\alpha = 2$) rows and by the n odd (for $\beta = 1$) or by the n even (for $\beta = 2$) columns of $\hat{B}$ and $\hat{C}$, respectively. In each case, the rows and columns are selected in the original order, as they follow in $\hat{B}$ and $\hat{C}$.

We will use several auxiliary matrices, the first of which is the matrix $A \in \mathbb{R}^{n\times n}$ given by $A_{ij} = -\sin[\frac{\pi}{n}(i-\frac{1}{2})j]$ for all $i, j \in \{1, ..., n\}$. (2)

Note: binary (yes-no) questions require justified answers (proofs) unless the contrary is explicitly stated.

(a) This problem is in part concerned with the verification of trigonometric equalities, and that work will be split among the students. Use the following table to identify your team on the basis of your name (as it is known to u:space).

The following table:

	last name starts with A-K	last name starts with L-Z
first name starts with A-K	Team 11	Team 12
first name starts with L-Z	Team 21	Team 22

Do not get carried away by the idea of teamwork: your solutions should be your own, and teamwork should take place at the exercise-group meeting .

(b) Let $J \in \mathbb{R}^{n\times n}$ be the permutation matrix associated with the permutation $(n, n-1, ..., 2, 1)$ of $\{1, ..., n\}$ see Example II.4.40 and Definition II.4.42 in the lecture notes. Express the matrix J entrywise (with ellipses) in the square-bracket notation (see the display equation in Remark II.4.8 for an example).

(c) Is the matrix J invertible? If yes, find its inverse.

(d) Give explicit expressions for the entries of the matrices $J\tilde{B}_{21}$ and $J\tilde{B}_{22}$. Give explicit expressions for the entries of the matrices $J\tilde{C}_{21}$ and $J\tilde{C}_{22}$.

(e) Let $S \in \mathbb{R}^{n\times n}$ be a matrix of downward one-position shift: $S \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ x_1 \\ \vdots \\ x_{n-2} \\ x_{n-1} \end{bmatrix}$ for each $x = [x_j]_{j=1}^n \in \mathbb{R}^n$. Is such a matrix S unique or not? Find one such a matrix S and express it entrywise (with ellipses) in the square-bracket notation (see the display equation in Remark II.4.8 for an example) .

(f) Is the matrix S you found in part (e) a permutation matrix? If yes, give the associated permutation.

(g) Is the matrix S you found in part (e) invertible? If yes, find its inverse.

(h) Give explicit expressions for the entries of the matrices CS and $AS^\top$.

(i) Of the following equalities, prove the one assigned to your team. (Team 11) $\tilde{C}_{11} = C$ (Team 21) $J\tilde{C}_{21} = C$ (Team 12) $\tilde{C}_{12} = B$ (Team 22) $J\tilde{C}_{22} = -B$

(j) Of the following equalities, prove the one assigned to your team, with suitable diagonal matrices $X, Y \in \mathbb{R}^{n\times n}$. (Team 21) $\tilde{B}_{11} = XC + YAS^\top$ (Team 11) $J\tilde{B}_{21} = YC - XAS^\top$ (Team 22) $\tilde{B}_{12} = XCS - YA$ (Team 12) $J\tilde{B}_{22} = YCS + XA$

Hint. Let us look at the right-hand side matrices of the above equalities. Note that, due to (1) and (2), for every $j \in \{1, ..., n\}$, the entries in the \$j\$th columns of C and A are cosines and sines of multiples of $j-1$ and j, respectively (by the way, think what similar statement you can make regarding $AS^\top$ and CS). So you may want to use trigonometric identities (see Problem 6 for a sufficient set of such identities) to express the entries in the \$j\$th columns of the left-hand side matrices from the above equalities as trigonometric functions of $(*) + (**)$, where $(*)$ denotes multiples of $j-1$ (for $\tilde{B}_{11}$ and $J\tilde{B}_{21}$) or of j (for $\tilde{B}_{12}$ and $J\tilde{B}_{22}$) and $(**)$ denotes quantities independent of j, and then apply suitable identities to split $(*)$ and \$(**) \$.

(k) Verify all the four equalities of part (i) in Julia for $n = 100$ and for $n = 101$.

(l) Using the matrices X and Y that you found in (your assigned equality of) part (j), verify all the four equalities of part (j) in Julia for $n = 100$ and for $n = 101$.

Side remark: high-school mathematics establishes a bad reputation for trigonometry, but see what a symmetry in $\hat{B}$ it reveals here!

1.2 Notes

1.3 Solution

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2 Problem 8

2.1 Task

Permutation matrices

(a) Let $n \in \mathbb{N}$ and $\Pi \in \mathbb{F}^{n \times n}$ be a permutation matrix. In terms of matrix multiplication, prove that there exist unique indices $\pi_k \in \{k, ..., n\}$ with $k \in \{1, ..., n\}$ such that $\Pi = \Pi_n \cdots \Pi_1$ where each Π_k with $k \in \{1, ..., n\}$ is the (k, π_k) -exchange matrix of order n (see Definition II.4.44).

(b) Count the number of permutation matrices of order $n \in \mathbb{N}$.

2.2 Notes

2.3 Solution

3 Problem 9

3.1 Task

Small data and big data

Let $n \in \mathbb{N}$ be such that $n \geq 2025$. For each of the following matrices, find a rank-r factorization with the least possible $r \in \mathbb{N}$ and prove r to be the least possible. Hint: to show the minimality of a factorization rank $r \in \mathbb{N}$, assume that a factorization of rank $r - 1$ exists and arrive at a contradiction.

(a) $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$

(b) $A = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$

(c) $A \in \mathbb{R}^{n \times n}$ given by $A_{ij} = i$ for all $i, j \in \{1, ..., n\}$.

(d) $A \in \mathbb{R}^{n \times n}$ given by $A_{ij} = i - j$ for all \$i,j∈\1,. . . ,n\.\$

(e) $A \in \mathbb{R}^{n \times n}$ given by $A_{ij} = i^2 + \cos \frac{\pi j}{n} - \cos \frac{\pi i}{n}$ for all $i, j \in \{1, ..., n\}$

3.2 Notes

3.3 Solution